# 5. Use Cases and Business Impact

This section describes how the project is intended to be used, which users it serves, and what business value it can and cannot provide. The use cases come from the project brief. For each one, this section states what the system does, how it helps, and what limits its usefulness.

## 5.1 Overview

The brief defines three business use cases:

1. Empower real estate investors with intelligent tools to assess long-term returns.
2. Support buyers in choosing high-return properties in developing areas.
3. Help real estate companies automate investment analysis for listings.

The Streamlit application provides three features that support these use cases:

| App feature | What it does |
|---|---|
| **Predict tab** | Analyses one property. Returns whether it is a Good Investment (with a confidence score), the estimated price after 5 years, the expected gain, and a comparison with other properties in the same city and BHK category. |
| **Explore tab** | Filters the data by state, BHK and price. Shows average price per sq ft by city, the Good Investment rate by property type, and a heatmap of price per sq ft by city and BHK. |
| **Model Performance tab** | Shows the metrics, confusion matrix, feature importance and EDA charts for transparency. |

## 5.2 Use case 1: Investors assessing long-term returns

**Who:** Individual investors and small investment groups who are deciding whether a property is worth buying to hold for several years.

**How the system helps:**

- The Predict tab gives a quick first screen. An investor can enter the details of a property and see whether it is cheap for its city, based on price per sq ft.
- The 5-year price estimate gives a simple view of possible value growth, using city-level growth rates.
- The peer comparison shows whether the property's price per sq ft is above or below the median for its city and BHK category.

**Limits:**

- The Good Investment label is based on price per sq ft relative to the city median. It does not measure rental income, resale history, or actual returns.
- The 5-year price uses assumed growth rates, not historical market data (see Section 4.3). It should not be used as a forecast.
- The classifier is correct about 71% of the time on test data. It should be treated as a screening aid, not a decision tool.

**Appropriate use:** The app can help an investor shortlist properties for further research. It should always be used with local market knowledge, legal checks and professional advice.

### 5.3 Use case 2: Buyers choosing high-return properties in developing areas

**Who:** Home buyers who want to find properties that may gain value, especially in areas that are still developing.

**How the system helps:**

- The Explore tab allows filtering by state, BHK and price, so a buyer can compare price levels across cities.
- The heatmap of price per sq ft by city and BHK helps identify cities where a given property size is relatively cheap.

**Limits:**

- The dataset has no information about whether an area is developing. There is no infrastructure score, no planned-development data, and no information about new transport links or business parks. The brief mentions these factors, but they are not in the data (see Section 4.5).
- The Explore tab does not filter by locality, so a buyer cannot compare neighbourhoods within a city.
- The dataset shows almost no difference in price per sq ft between cities or localities (Section 2, Q6 to Q10). The data cannot show which areas are cheaper or growing faster.

**Appropriate use:** The app can show whether a specific property is priced low for its city. It cannot identify developing areas. This use case needs additional data on infrastructure, planned projects and historical price growth before it can be supported.

### 5.4 Use case 3: Real estate companies automating investment analysis for listings

**Who:** Property portals, brokerages and developers who publish many listings and want to tag them with an investment indicator.

**How the system helps:**

- The trained classifier and regressor are saved as files (`models/classifier.joblib` and `models/regressor.joblib`) and can be loaded in any Python environment.

- The same prediction function used in the app (`predict_property`) can be run in a loop over a list of listings to produce a Good Investment flag and a 5-year price estimate for each one.
- Batch processing takes only a few seconds per thousand listings on a standard machine, based on the prediction speed of the Random Forest models.

**Limits:**

- Listings need the same input fields as the training data, including Price, Size, City, BHK, Year_Built and Amenities. Missing fields are filled with median or most-frequent values, which may reduce accuracy.
- Category values in the listings must match the dataset categories (for example, Owner_Type values Owner, Builder and Broker). Other values are treated as unknown.
- The model reflects the patterns in this dataset. If a company's listings come from a different market, the predictions may not be reliable without retraining on that company's data.

**Appropriate use:** A company could use the model to flag listings for internal review, for example to highlight properties priced low relative to their city. It should not use the flag as the only basis for pricing or for advice to customers.

### 5.5 Business impact assessment

The project was built as an academic demonstration. The following table summarises the potential impact, and what would be needed to measure it properly.

| Area | Potential impact | Evidence available now | What is needed to measure it |
|---|---|---|---|
| Investor screening | Faster first-pass review of many properties | Predict tab works, classifier ~71% accurate on test data | User testing with real investors, comparison with expert screening |
| Buyer decisions | Quick check of whether a price is high or low for a city and BHK | Peer comparison in Predict tab, Explore heatmap | Real market price data, and evidence that buyers use the tool |
| Company automation | Automatic tagging of listings | Batch-ready model files, prediction function | Trial on real listings, measurement of time saved and error rate |
| Return assessment | Estimate of long-term value | 5-year estimate using assumed growth rates | Real historical price data for each city and locality |

**Summary of business value:** In its current form, the project shows that a simple, explainable model can be built and deployed. It can support early screening of properties. Its main business value would come from using real market data, which is the most important step before any commercial use.

## 5.6 Risks and responsible use

| Risk | Description | Mitigation |
|---|---|---|
| Over-reliance on the label | Users may treat "Good Investment" as financial advice | Clear disclaimer in the app and documentation (see Section 4.10) |
| Bias toward larger properties | The classifier depends on size (Section 4.4) | Document the dependence, and test the model without size before any real use |
| False growth expectations | The 5-year estimate may be read as a real forecast | Label the output as based on assumed growth rates |
| Errors on real data | Model accuracy on real listings is unknown | Validate on real listings with known outcomes before use |
| Data privacy | Real listings may contain personal information | Use only property features, not seller or buyer details |

## 5.7 Recommended next steps

If this project were developed further for real use, the following steps would be needed, in order of priority:

1. **Get real market data:** historical sale prices by city and locality, with sale dates.
2. **Add the missing features:** infrastructure, planned development, crime data and transport links, as described in the brief.
3. **Replace the rule-based labels:** define Good Investment using actual returns (for example, price growth over several years after accounting for costs).
4. **Remove the size dependence test:** check model accuracy without Size_in_SqFt, and report the result.
5. **Validate with a three-way split:** separate training, validation and test sets, to avoid test-set selection bias.
6. **Get expert review:** have a property analyst or economist review the features, labels and assumptions.

These steps would turn the academic prototype into a tool that could support real investment decisions.